

AP PHYSICS C: ELECTRICITY AND MAGNETISM NOTES

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1 Circuits

Equipment:

- D-cell battery/cells: with positive and negative terminals.
- Battery holder.
- Ammeter (machine labeled “A”).
- Multimeter (can be ammeter, voltmeter or ohmmeter): has “test leads” or “probes.”
- Single pole single throw switch and single pole double throw switch.
- Wires: alligator-alligator wires (alligator clips on both ends), alligator-banana wires, banana-banana wires.
- Sockets (light bulbs screw into the hole).
- Round bulbs and long bulbs.

There are electrons everywhere in the circuit (all metal solids have delocalized electrons). The battery serves as a “pump” to move electrons around through the circuit. We can use the voltmeter in the multimeter to measure the “pump strength.”

We can find the potential difference across the battery $\Delta V_{\text{battery}}$ and across the two light bulbs ΔV_{B_1} and ΔV_{B_2} .

$$\Delta V_{\text{battery}} = 4.34 \text{ V}$$

$$\Delta V_{B_1} = 1.84 \text{ V}$$

$$\Delta V_{B_2} = 1.83 \text{ V}$$

Since energy is conserved the potential difference of the bulbs should be the same as that of the battery, but it is not. We need to take ΔV of the wires and switch into account as well.

Potential difference (ΔV) is measured in volts (V), and is related to energy. In Ye Olde Times, it is denoted V or emf (electromotive force), and ΔV_{bat} is denoted ε .

Flow rate/current (I) measures the number of electrons that pass a certain point per unit of time, and is measured in amperes (A).

Resistance (R) is how difficult it is for electrons to flow through and is measured in ohms (Ω).

The current is equal across the circuit.

Ohm's Law

$$\Delta V = IR$$

where:

ΔV = potential difference (V)

I = current (A)

R = resistance (Ω)

Resistors convert electrical energy to thermal energy. Light bulbs are a special case of resistor that convert electrical energy to thermal and light energy.

Resistors in series

When resistors are connected in series, current is equal across any part of the circuit:

$$I_{\text{bat}} = I_1 = I_2 = I_3$$

Assuming no potential is lost anywhere:

$$\Delta V_{\text{bat}} = \Delta V_1 + \Delta V_2 + \Delta V_3$$

$$R_{\text{total}} = R_1 + R_2 + R_3$$

Resistance

$$R = \frac{\rho l}{A}$$

where:

ρ = resistivity of material ($\Omega \text{ m}$)

l = length (m)

A = cross-sectional area (m^2)

A **junction/node** is a point where three or more wires are connected.

A **branch** connects two nodes.

To connect a parallel circuit, multiple junctions need to be made.

Resistors in parallel

The currents are different at different parts of the circuit:

$$\begin{aligned} I_{\text{bat}} &= I_1 + I_2 + I_3 \\ \Delta V_{\text{bat}} &= \Delta V_1 = \Delta V_2 = \Delta V_3 \\ \frac{1}{R_{\text{total}}} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \end{aligned}$$

Similar to springs in parallel.

A shortcut: if the total equivalent resistance of a parallel circuit is R_T and the current entering the circuit is I_T , the current through branch i is. Since the circuit is parallel R_T is less than the individual R_i 's.

$$I_i = I_T \cdot \frac{\frac{1}{R_i}}{\sum_k \frac{1}{R_k}} = I_T \cdot \frac{\frac{1}{R_i}}{\frac{1}{R_T}} = I_T \cdot \frac{R_T}{R_i}$$

From $I = \Delta V/R$ and $\Delta V_{\text{bat}} = \Delta V_1 = \Delta V_2 = \Delta V_3$:

$$I_{\text{bat}} = I_1 + I_2 + I_3 \implies \frac{\Delta V_{\text{bat}}}{R_{\text{total}}} = \frac{\Delta V_1}{R_1} + \frac{\Delta V_2}{R_2} + \frac{\Delta V_3}{R_3} \implies \boxed{\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

Electron Sea Model

In metals, atoms form metallic bonds: each metal atom releases its valence electrons, which then form a sea of delocalized electrons. These electrons are then attracted to multiple atoms. These bonds are very strong and lead to high melting points for metals.

Delocalized electrons can move in a direction caused by an electric force.

Electrons flow from the negative terminal to the positive terminal. Conventional current is the direction in which positive charges flow, and is opposite to the direction electrons flow. The direction electrons flow is marked e^- and conventional current is marked I . The “long” end of the battery symbol is positive.

Ohmmeters are connected when there are no charges flowing; voltmeter and ammeter need the circuit to be

active. Ammeter is connected in a circuit; voltmeter and ohmmeter measure across a circuit component (with a multimeter). In reality, these devices will modify the circuit when they are attached. An ammeter actually has a very small resistance, but we assume it has zero resistance. Also assume that a voltmeter has infinite resistance.

If two components are on different paths to the battery, they have the same potential difference (parallel).

A **combination circuit** is one which is not purely series or purely parallel. We can analyze such a circuit by calculating the total/equivalent resistance.

Power

$$P = I\Delta V = I^2 R = \frac{\Delta V^2}{R}$$

where:

P = power (W)

I = current (A)

ΔV = potential difference (V)

R = resistance (Ω)

The sum of power in each resistors equals the power in the battery.

Resistors dissipate energy by converting electrical energy to thermal energy at a rate equal to $P = I\Delta V = \dots$

Light bulb ratings

Suppose a light bulb is rated at p W and v V. p relates to the brightness of the light bulb. From the formula $P = (\Delta V)^2/R$, we can determine the resistance of the light bulb: $R = v^2/p$.

Light bulb assumptions:

- The brightness of a light bulb is directly proportional to the power.
- The resistance of a light bulb is constant.

Electric bills

The house is billed by energy used in kWh, which is actually a unit of energy.

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

In a short circuit, a path with small/virtually zero resistance forms, adding a parallel path. The overall/equivalent resistance drops even though ΔV_{bat} is constant, increasing the current and therefore the power. A circuit breaker/fuse will trip/melt and open the circuit when current is higher than a certain limit.

1.1 Internal Resistance

In an ideal battery, there is no resistance: $R = 0$. In practice, a battery contains chemical reactions to generate a potential difference/current, and as the products of the reaction increase, the internal resistance increase.

In a circuit diagram, a resistor is placed to either side of the symbol for a battery. Sometimes, a dotted line

includes these components. The potential difference of the battery and its internal resistance is ΔV_T (V).

Potential difference and internal resistance of battery

$$\Delta V_T = \varepsilon - Ir$$

where:

ΔV_T = terminal potential difference (includes internal resistance) (V)

ε = potential difference of ideal battery (V)

I = current through battery (A)

r = internal resistance (Ω)

When batteries are connected in series, the ideal potential difference (ε) increases, but internal resistance also increases. The actual increase in terminal potential difference (ΔV_T) might be quite low, and a lot of thermal energy is produced.

If batteries are connected in parallel, there is no increase in potential difference. But the current in the branches where batteries are connected is reduced as batteries are added in parallel, which reduces internal resistance and ΔV due to internal resistance Ir . The battery also lasts longer.

Potential difference opposing each other: positive terminals are adjacent to each other. The battery with a lower potential difference is being “charged,” until the batteries have the same potential difference.

Kirchhoff's Laws

1. Junction/node law (law of conservation of charges): at any junction (place where 3 or more wires are connected together), the sum of all currents flowing into the junction equals the sum of all currents flowing out of the junction. Charges are conserved.
2. Loop law: around any loop in the circuit, the sum of all ΔV changes must equal zero. In conventional current: the battery produces a positive ΔV , resistors and light bulbs produce a negative ΔV . The sum of all components equals 0. Opposite of using flow of electrons as direction.

The **ground** is where potential is zero volts. It is an ideal infinite source or sink of charge that can absorb unlimited charge without changing its potential of zero. It has both positive and negative charges.

1.2 Ideal Assumptions

- An ideal wire has zero resistance.
- An ideal ammeter has zero resistance.
- An ideal voltmeter has infinite resistance.
- An ideal battery has zero internal resistance.
- An ideal resistor is ohmic: it follows Ohm's law in that potential difference (ΔV) is directly proportional to current (I), with the slope being resistance R .

A light bulb is not ohmic: its temperature will increase, increasing its resistance.

1.3 Charge

Current is the flow of charged particles, the charge (q, Q) being measured in coulombs (C). One coulomb is the amount of charge delivered by a current of one ampere per second.

$$\left[A = \frac{C}{s} \right] \iff [C = A \cdot s]$$

An electron has a charge of -1.6×10^{-19} C, and the charge of a proton has the same magnitude but opposite sign: $+1.6 \times 10^{-19}$ C.

Charge and current

$$I = \frac{Q}{t} = \frac{dq}{dt}$$

where:

I = current (A)

Q = charge (C)

t = time (s)

This is helpful in non-steady state circuits where current is not constant and changes over time.

2 Electrostatics

Coulomb's law

$$F_E = \frac{kq_1q_2}{r^2} \quad k = \frac{1}{4\pi\epsilon_0}$$

where:

F = electrostatic force (N)

k = Coulomb constant

$$= 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$$

ϵ_0 = permittivity of free space (C^2/Nm^2)

$$= 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

q_1, q_2 = charge (C)

r = distance between particles (m)

There are positive and negative charges. The magnitude of the force is:

$$\|F_E\| = \frac{k|q_1||q_2|}{r^2}$$

from which the direction of the force can be figured out by logic: opposite charges attract.

A **Van de Graaff generator** generates electrostatic energy by using a belt to transport electrons to a spherical dome. Electrons try to spread out as much as possible and are positioned on the dome's surface.

Constants (on equation sheet)

The mass of an electron is:

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

The elementary charge is the smallest possible charge, and is equal to the magnitude of the charge of a proton or electron.

$$Q_e = 1.60 \times 10^{-19} \text{ C}$$

The gravitational constant from Mechanics is:

$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

SI prefixes: 10^{-3} milli (m), 10^{-6} micro (μ), 10^{-9} nano (n), 10^{-12} pico (p).

For two electrons spaced 1 millimeter apart, we can calculate $F_E = k |q_1| |q_2| / r^2$ and $F_G = G m_1 m_2 / r^2$. It turns out that F_E is about 4×10^{42} times greater than F_G . So the gravitational force can often be ignored because it is so weak compared to electric force.

Electric force is a vector quantity, whose direction can be indicated with $+/-$ sign or with $\hat{i}, \hat{j}, \hat{k}$.

Electrostatics vs circuits

In electrostatics, charges are not moving (hence “static”). In a circuit, which for our purposes is direct current (DC), charges are flowing through the circuit and are not stationary.

2.1 Atomic Structure

Matter is made of atoms, ions and molecules, which are in turn made of protons (with positive charge), electrons (with negative charge) and neutrons (with no charge).

The mass of an electron is significantly lower than the mass of a proton or neutron:

$$(m_e = 9.11 \times 10^{-31} \text{ kg}) \ll m_p = m_n = 1.67 \times 10^{-27} \text{ kg}$$

Charge is quantized: the magnitude of charge on a proton or electron is the elementary charge:

$$|Q_e| = |Q_p| = 1.6 \times 10^{-19} \text{ C}$$

There is no unit of charge smaller than the elementary charge, and all possible charges is a multiple of it:

$$|Q| = n (1.6 \times 10^{-19}) \text{ C} \quad n \in \mathbb{Z}^+$$

An object being “charged” indicates the sign of the net charge, which is either positive or negative.

- Positively charged: the object is deficient of electrons (has less electrons than protons).
- Negatively charged: the object is in excess of electrons (has more electrons than protons).

The mass of excess electrons in a negatively charged object or the mass of electrons removed in a positively charged object is negligible, because the mass of an electron is so low.

Charges on a metal/conductive sphere (solid or hollow) will always be on the surface of the sphere, because charges tend to maximize distance between each other. Thus, the net charge is always on the surface.

In an insulator, there is no free flow of charges. Charges are stationary on the surface of the object.

2.2 Polarization

Polarization is a separation of positive and negative charges within an object. This is done by shifting the statistical distribution of electrons to one side of an object (as explained by quantum mechanics, electron orbitals, etc.). It does not create charge, and will still have the same net charge as before.

A charged object can induce polarization in nearby neutral objects. Once the neutral object is polarized, like charges repel and are now further apart, meaning opposite charges are now closer together. Therefore, the two objects will attract.

In conductors, electrons move more readily. Polarization results in a side with excess electrons (negative) and a side with a deficit in electrons (positive).

In insulators, electrons are not free to move, and polarization occurs by changing the shape of electron clouds/orbitals. The electron cloud becomes asymmetrically shaped, and the “center of electrons/charge” is not at the atomic nucleus. The individual atoms become polarized.

Polar molecules (have a dipole moment) can orient themselves in an electric field due to attraction.

Polarization explains a number of real-world occurrences, such as:

- Balloons stick to a wall because the balloon is negative charged (has high affinity for electrons), and the wall molecules polarize.
- Water streams bend because water molecules reorient toward a charged object.
- Small pieces of paper jump toward a balloon because the negatively charged balloon surface polarizes it.

2.3 Charging

Charging by friction. Different objects (insulators) have a different tendency to gain or lose net charge when rubbed with another object. Some will gain electrons while some will lose electrons.

- Glass and human hair will lose electrons and become more positive.
- Nylon and wood will gain electrons and become more negative.

Charging by conduction. This can be used for both insulators and conductors. It involves the movement of charges to/from a neutral conductor from/to a charged object by contact.

Charging by induction.

1. Start with a charged object (“rod”) and a neutral conductor (“sphere”).

2. When the rod is brought near (not necessarily contacting) the sphere, free electrons will rush towards/away from the rod. The sphere becomes polarized.
3. Connect the sphere to ground.
4. Electrons from ground will rush towards the sphere, and displace the positively charged particles.

OR

Electrons from the sphere will rush towards ground, leaving only positively charged particles.

5. When the ground wire is removed from the sphere and the rod is moved away, the sphere remains charged.

This can be used to charge the sphere with either positive charge or negative charge. It would depend on the charge of the rod we start with.

- Negatively charged rod $\rightarrow$ positively charged sphere.
- Positively charged rod $\rightarrow$ negatively charged sphere.

Electroscopes. An electroscope is a device used to detect charge. It consists of a large model knob connected via a stem to two metal leaves. If a charged object is brought near the knob, electrons will move towards/away from the leaves, causing them to move upwards (since the leaves become charged and repel each other).

2.4 Field Model

The field model is related to non-contact forces, which are gravity, electric force and magnetic force. The field model describes the alteration of space.

Gravity

In a gravitational field, all masses are attracted to the center (of mass M) by a force given by the law of universal gravitation:

$$F_G = \frac{GMm}{r^2}$$

The gravitational field strength g is:

$$g = \frac{GM}{r^2}$$

and has units $\text{m/s}^2 = \text{N/kg}$. g is a vector quantity depending on the distance vector r and the location of the test mass. The force exerted on a mass m is:

$$F = gm \implies \Sigma F = ma$$

Field lines all point towards the mass M . The field is radially symmetric and the field lines point radially inward. On the Earth's surface, field lines point down and is uniform at $g = 9.81 \text{ m/s}^2$.

Electricity

In an electric field, charges are either attracted to or repelled from the central charge (Q), given by:

$$F_E = \frac{kQq}{r^2}$$

The electric field strength E is:

$$E = \frac{kQ}{r^2}$$

and has units N/C . E is a vector quantity depending on the distance vector r and the location of the test charge. The force exerted on a charge q is:

$$\Sigma F = Eq$$

Field lines are more complex because there are both attractive and repulsive force. The direction of field lines are the direction of F_E on a positive charge. When close to Q , the field lines are approximately straight and are uniform.

2.5 Electric Fields

A **capacitor** is made of two plates with opposite charge. If the distance between the plates is small, it can be thought of as being uniform.

A field is **uniform** if the electric force on a test charge is the same (in both direction and magnitude) at all points in space. A field is **nonuniform** if it is not.

Properties of electric fields:

- Electric field is always a vector quantity around the source charge (Q). At any location, the direction of the electric field is tangent to field lines. Field lines can be curved.
- Field lines are perpendicular to conducting surfaces.
- Field lines point in the direction of a force on a positive
- The density/spacing of field lines indicate the relative strength of the field. Denser lines/lower distance between lines indicate a stronger electric field. If the strength of the electric field $E = kQ/r^2$ doubles, the number of lines doubles.
- Field lines do not cross each other, as this would indicate two different electric field values/forces at the same point (the electric field is not a function).
- Electric field strength E is a vector quantity.

Electric field strength

The strength of an electric field created by a point charge with charge Q from a certain distance r is:

$$\|E\| = \frac{k|Q|}{r^2}$$

The units for electric field strength E is either newton per coulomb (N/C) or volt per meter (V/m). This is not dependent on the strength of any test charge. The force acting on a test charge of charge q is:

$$\|F_E\| = E|q| = \frac{k|Q||q|}{r^2}$$

2.6 Electrical Potential and Potential Energy

In a uniform gravitational field, change in potential energy is:

$$\Delta U_g = mg\Delta h$$

where g is the gravitational field strength, which is 9.81 m/s^2 on the surface of the Earth.

Gravitational potential

Gravitational potential is gravitational potential energy per unit mass:

$$V = \frac{U_g}{m} = \frac{mgh}{m} = gh$$

and is measured m^2/s^2 or J/kg.

This concept can be similarly used for electric potential.

Electric potential and potential energy in a uniform field

In a uniform electric field (between two charged plates) of strength E , the **electric potential energy** U_E of a charge q at a distance x to the plate of opposite charge from q is:

$$\Delta U_E = qEx = q\Delta V$$

Electric potential is potential energy per unit charge:

$$V = \frac{U_E}{q} = \frac{qEx}{q} = Ex = \frac{kQ}{x}$$

Change in potential energy associated with a change in potential is:

$$|\Delta U_E| = \Delta |q \underbrace{Ex}_V| = \Delta |qV| = |q| |\Delta V|$$

Equipotential lines/surfaces are lines/surfaces on which electric potential is constant. They are described in terms of the electric potential on all points on the line/surface measured in volts (V). They behave similarly to contours.

- Potential is highest when close to a charge, and decreases as distance r, x increases.
- Positive point charges have positive potential and negative point charges have negative potential. For both positive and negative charges, electric potential at infinite distance is 0 V.
- Positive charges move from high to low potential values.
- Electric field lines/surfaces are perpendicular to equipotential lines.
- Field strength is inversely proportional to the spacing between field lines.
- Electric field is a vector quantity, and electric potential is a scalar quantity.

If a point P is affected by multiple charges, the electric potential at P is the sum of the electric potential created by each charge:

$$V_i = \frac{k|Q_i|}{x_i} \quad V_P = \sum_i V_i$$

Electric potential is measured in volts (V). From this equation:

$$\left[V = \frac{\text{N/C}}{\text{m}} = \frac{\text{Jm}}{\text{Cm}} = \frac{\text{J}}{\text{C}} \right]$$

If the potential difference ΔV is known, change in electric potential energy ΔU_E can be found. Kinetic energy K can also be found: since energy is conserved, $\Delta K = -\Delta U_E$.

2.7 Energy

In mechanics, the unit for energy is the joule (J), which is defined as:

$$\left[\text{J} = \text{N m} = \frac{\text{kgm}^2}{\text{s}^2} \right]$$

Energy can also be measured in electron-volts (eV). The change in energy of a proton or electron (with the same charge $q = 1.6 \times 10^{-19} \text{ C}$) is given by:

$$|\Delta U_E| = |q| \cdot \Delta V = 1.6 \times 10^{-19} \cdot 1 = 1.6 \times 10^{-19} \text{ J}$$

A **electron-volt** is equal to the kinetic energy gained by an elementary charge (proton or electron) as it accelerates across a potential difference of 1 V.

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

The energy of assembly is the energy needed to construct a system of point charges.

1. The energy it takes to add the first point charge is zero, because there is zero potential initially and zero potential with only one charge.
2. The change in energy to add a second charge is:

$$\Delta E = q\Delta V = q_2 \left(\frac{kq_1}{d} - 0 \right) = \frac{kq_1q_2}{d}$$

The second charge starts at “infinity” so it has a potential of 0.

3. The change in energy to add a third charge is:

$$\Delta E = q_3 \left(\frac{kq_1}{d} + \frac{kq_2}{d} - 0 \right) = \frac{kq_3(q_1 + q_2)}{d}$$

The total energy to construct the system is:

$$\text{Total Energy of Assembly} = \frac{kq_1q_2}{d} + \frac{kq_1q_3}{d} + \frac{kq_2q_3}{d}$$

Electric potential energy of a system

The total potential energy of a system, which, like gravitational energy, is the sum of the potential energies generated by each combination of two charges. The actual charges are substituted, not only their magnitudes.

$$U_E = \sum \frac{kq_1q_2}{r}$$

Two like charges tend to move apart because they have high potential energy when held together, and moving away reduces potential energy. Two opposite charges attract because they have lower potential energy (negative sign, but high magnitude) when close together.

Electric potential energy U_E equals the total energy of assembly.

2.8 Electrostatics and Circuits

When a circuit is open electrons are moving in random directions at 10^6 m/s .

When a circuit is closed, the net movement of electrons is in a certain direction (direction of electron flow, opposite of conventional current). There are electrons moving in the opposite direction still. The speed of the net movement (“drift speed”) is around 10^{-5} to 10^{-4} m/s . Individual electrons are much faster though.

Current

Current (in Ohm's law as $I = \Delta V/R$) is the net flow of charge per unit time and is measured in amperes (A) or coulombs per second (C/s):

$$I = \frac{Q}{t} = \frac{\Delta Q}{\Delta t} = \frac{dq}{dt}$$

Current can also be defined as:

$$I = Ne v_d A$$

where:

N = conduction electron density (number of electron carriers per unit volume)

e = elementary charge (C)

$$= 1.6 \times 10^{-19} \text{ C}$$

v_d = drift velocity (m/s)

A = cross-sectional area (m^2)

Current $I = \Delta V/R$ is measured in amperes (A): $[A = V/\Omega]$.

Power $P = I\Delta V = I^2 R = \Delta V^2/R$ is measured in watts (W): $[W = J/s]$.

Resistivity $R = \rho l/A$ is measured in ohms (Ω): $[\Omega = V/A]$.

Conductivity

Conductivity σ measures a material's ability to conduct electricity:

$$\sigma = \frac{1}{\rho}$$

where:

σ = Conductivity ($1/\Omega\text{m}$)

ρ = Resistivity (Ωm)

Drift velocity v_d is the average displacement traveled by charge carriers per unit time, and as a measure of velocity is measured in m/s.

Current density

Current density is the current per unit area and is a vector quantity.

$$\vec{J} = \frac{I}{A}$$

Current can be expressed as an integral of current density:

$$I = \int \vec{J} dA \stackrel{?}{=} \vec{J} A$$

Electric field strength is:

$$\vec{E} = \rho \vec{J}$$

where:

J = current density ($\text{A}/\text{m}^2 = (\text{C}/\text{s})/\text{m}^2$)

I = current (A)

A = cross-sectional area (m^2)

E = electric field strength ($\text{N}/\text{C} = \text{V}/\text{m}$)

J and E are vector quantities (even though from $J = I/A$, it does not appear as such!).

2.9 Continuous Charge Distribution

An infinite number of infinitesimally small point charges (with charge dq) are distributed throughout an object. We want to determine the net electric field E_{net} at a location P .

<< and >>

If $a \ll b \iff b \gg a$, then a is small compared to b . Then any sum $a + b$ can be considered to be only b .

2.9.1 Thin rod, perpendicular point of interest

We want to find the net electric field at a point P located on the perpendicular bisector of the rod (the line from P to the center of the rod forms a right angle with the rod).

The rod has a total charge $+q$, and P is at a perpendicular distance y from the rod. Every section of the rod, each with charge dq and with dx , has an effect on the rod. Charges from sections on both sides (both on the left and right) of the rod will cancel out with each other, so the net electric field is directly away from the rod.

Linear charge density

The linear charge density λ of the rod is:

$$\lambda = \frac{q}{L} = \frac{dq}{dx} \iff dq = \lambda dx$$

From the definition of electric field, the electric field created by the rod section is:

$$E = \frac{kQ}{r^2} \implies dE = \frac{k dq}{r^2} = \frac{kq dx}{r^2}$$

We only care about the vertical component of the charge on P by each rod section (horizontal components cancel

out). The vertical component is:

$$dE_{\text{vertical}} = dE \cos \theta = \frac{kq \cos \theta}{r^2} dx$$

Integrating this over the rod (for each section) gives the net electric field. Distance r can be expressed in terms of x and y components of distance: $r = \sqrt{x^2 + y^2}$.

$$\begin{aligned} E_{\text{net}} &= \int dE_{\text{vertical}} = \int \frac{k\lambda \cos \theta}{r^2} dx = k\lambda \int \frac{\cos \theta}{r^2} dx = k\lambda \int_{-L/2}^{L/2} \frac{1}{(x^2 + y^2)^{3/2}} dx \\ &= \left[\dots \text{trig. substitutions} \dots \right] = k\lambda y \left[\frac{x}{y^2 \sqrt{x^2 + y^2}} \right]_{-L/2}^{L/2} = \boxed{\frac{2k\lambda L}{y \sqrt{L^2 + 4y^2}}} \end{aligned}$$

where:

k = Coulomb constant

λ = linear charge density (uniform/constant)

y = distance between rod and P

L = length of rod

Special case 1. If P is very far from the rod, then $y \gg L$ and $L/y \rightarrow 0$. Also, recall that $\lambda = q/L \Rightarrow q = \lambda L$.

$$E = \frac{2k\lambda L}{y \sqrt{L^2 + 4y^2}} = \frac{2k\lambda L}{2y^2} = \boxed{\frac{kq}{y^2}}$$

Special case 2. If the rod is infinitely long, then $L \gg y$

$$E = \frac{2k\lambda L}{y \sqrt{L^2 + 4y^2}} = \frac{2k\lambda L}{yL} = \boxed{\frac{2k\lambda}{y}}$$

Electric field due to infinitely long rod

The electric field due to an infinitely long rod with charge density λ on a point a distance r away is:

$$E = \frac{2k\lambda}{r}$$

Electric field strength E is directly proportional to the inverse of distance $1/r$.

2.9.2 Thin rod, colinear point of interest

A point P is located on the same line in space as the rod of length L and charge density λ , at a distance a from the closest end. (If the rod were to be infinite distance, P would be on the rod.)

As before, linear charge density gives $dq = \lambda dx$ and electric field strength $E = kQ/r^2$.

$$dE = \frac{k dq}{r^2} = \frac{k\lambda dx}{r^2} = \frac{k\lambda}{x^2} dx$$

The total charge is the integral of dE over the rod, which is bounded by $a \leq x \leq a + L$.

$$\begin{aligned} E &= \int_a^{a+L} dE = \int_a^{a+L} \frac{k\lambda}{x^2} dx = k\lambda \left[-\frac{1}{x} \right]_a^{a+L} = -k\lambda \left(\frac{1}{a+L} - \frac{1}{a} \right) \\ &= -k\lambda \cdot \frac{a - (a+L)}{a(a+L)} = k\lambda \cdot \frac{(a+L) - a}{a(a+L)} = \boxed{\frac{k\lambda L}{a(a+L)}} \end{aligned}$$

Special case. If P is infinitely far away from the rod, then $a \gg L$. Recall that $k\lambda L = q$.

$$E = \frac{k\lambda L}{a(a+L)} \overset{0}{=} \frac{k\lambda L}{a^2} = \boxed{\frac{kq}{a^2}}$$

This behaves the same as a point charge (for which $E = kq/r^2$).

2.9.3 Semicircular arc

Unlike the previous cases, the semicircular arc is negatively charged with net charge $-q$. It has radius r . P is a point at the center of the semicircle. dE is the electric field on P by a section of the arc with charge dq . The same section on the other side of the arc will cancel out its horizontal component. So, like before, the net electric field on P points straight up.

The charge on the semicircular arc is distributed along the arc as if it were one dimensional. The charge density λ is still q/L , where L is the length of the arc. s is the arc length, and ds is the length of the arc section.

$$L = \frac{1}{2}2\pi r = \pi r \implies \lambda = \frac{q}{L} = \frac{q}{\pi r} = \frac{dq}{ds} \implies dq = \lambda ds$$

Arc length $s = r\theta \implies ds = r d\theta$. The vertical component of the charge from the section is:

$$dE_{\text{vertical}} = dE \sin \theta = \frac{k dq}{r^2} \sin \theta = \frac{k\lambda \sin \theta}{r^2} ds = \frac{k\lambda \sin \theta}{r^2} r d\theta = \frac{k\lambda \sin \theta}{r} d\theta$$

Integrating this over the arc gives the net electric field. Because this is a semicircular arc, $0 \leq \theta \leq \pi$, which are the bounds of integration.

$$\begin{aligned} E &= \int_0^\pi dE_{\text{vertical}} = \int_0^\pi \frac{k\lambda \sin \theta}{r} d\theta = \frac{k\lambda}{r} \int_0^\pi \sin \theta d\theta = \frac{k\lambda}{r} [-\cos \theta]_0^\pi \\ &= -\frac{k\lambda}{r} (\cos \pi - \cos 0) = -\frac{k\lambda}{r} (-1 - 1) = -(-2)\frac{k\lambda}{r} = \boxed{\frac{2k\lambda}{r}} \end{aligned}$$

In general, the electric field of any circular arc is:

$$E = \frac{k\lambda}{r} [\sin \theta]_{-\theta}^\theta$$

where $[-\theta, \theta]$ are the bounds of the arc.

2.9.4 Circle of charge

The effect of a circle of charge on a point lying on the line through the center of the circle, and forming a right angle with the circle. The distance between the center of the circle and the point is z and the circle has radius R .

A small section of the circle of charge has charge dq and arc length ds . r is the distance between the section and the point of interest, forming an angle θ .

Linear charge density of the charge section is:

$$\lambda = \frac{q}{L} = \frac{dq}{ds} \implies dq = \lambda ds$$

The electric field from the charge section is:

$$E = \frac{kq}{r^2} \implies dE = \frac{k dq}{r^2} = \frac{k\lambda ds}{r^2}$$

The vertical components of dE cancel, so we are only interested in the horizontal component. From the diagram, r is the hypotenuse: $r^2 = R^2 + z^2$ and $\cos \theta = z/R$.

$$dE_{\text{horizontal}} = dE \cos \theta = \frac{k\lambda ds}{r^2} \cos \theta = \frac{k\lambda ds}{z^2 + R^2} \cdot \frac{z}{r} = \frac{k\lambda ds}{z^2 + R^2} \cdot \frac{z}{(z^2 + R^2)^{1/2}} = \frac{k\lambda z ds}{(z^2 + R^2)^{3/2}}$$

The net charge is the integral of $dE_{\text{horizontal}}$, but everything is constant except ds . We integrate over the circumference of the circle: $0 \leq s \leq 2\pi R$.

$$E = \int \frac{k\lambda z}{(z^2 + R^2)^{3/2}} ds = \frac{k\lambda z}{(z^2 + R^2)^{3/2}} \int_0^{2\pi R} ds = \frac{k\lambda z}{(z^2 + R^2)^{3/2}} \cdot 2\pi R = \boxed{\frac{kqz}{(z^2 + R^2)^{3/2}}}$$

Special case 1. If P is at the center of the circle of charge, then $z = 0 \implies E = 0$.

Special case 2. If P is very far from the ring, $z \gg R$.

$$E = \frac{kqz}{(z^2 + R^2)^{3/2}} \stackrel{0}{\approx} \frac{kqz}{z^3} = \frac{kq}{z^2}$$

When the distance is high compared to the radius of the circle, the ring of charge becomes a point charge.

2.9.5 Solid sphere or hollow shell

The sphere/shell has radius R .

Inside. When inside the shell, sphere or any conductor, the net electric charge is 0.

On the surface. The net electric field is:

$$E = \frac{kQ}{R^2}$$

Away from the surface. The net electric field is:

$$E = \frac{kQ}{r^2}$$

where r is the distance from the point of interest to the center of the sphere/shell.

When on/away from the surface, entire sphere/shell can be thought of as being a point charge.

2.10 Gauss's Law

We can consider electric field lines/vectors as a flowing fluid (such as a gas or liquid).

An **area vector** is a vector representing a surface:

- Its length is proportional to the magnitude of the area.
- Perpendicular to the surface area it represents (assuming the surface is small enough to be a plane).
- Since electric field E is perpendicular to a conducting surface, E at a point is a scalar multiple of dA .

Flux

Flux (Φ) is the flow of a fluid through a cross-sectional area (CSA, A).

$$\Phi = \mathbf{v} \cdot \mathbf{A} = vA \cos \theta$$

where:

$\mathbf{v}$ = velocity of liquid

A = cross-sectional area vector

$v = \|\mathbf{v}\|$ = speed of liquid (m/s)

$A = \|\mathbf{A}\|$ = cross-sectional area (m)

θ = angle between flow vector and area vector (rad)

Flux is the volume flow rate and is measured in m^3/s . Flux is a scalar.

Electric flux

Electric flux is the amount of electric field passing through a surface area. Electric flux is given by:

$$\Phi_E = \mathbf{E} \cdot \mathbf{A} = EA \cos \theta$$

where:

$\mathbf{E}$ = electric field

A = cross-sectional area vector

$E = \|\mathbf{E}\|$ = strength/magnitude of electric field (N/C, V/m)

$A = \|\mathbf{A}\|$ = cross-sectional area (m)

θ = angle between electric field vector and area vector (rad)

Suppose we know the electric field $d\mathbf{E}$ created by a small section of a surface with area vector $d\mathbf{A}$.

$$dE = \mathbf{E} \cdot d\mathbf{A}$$

We integrate the electric field over the surface for the total electric field. We integrate over a surface, so we use the **surface integral**.

$$E = \oint \mathbf{E} \cdot d\mathbf{A}$$

If $Q_{\text{enc}} = 0$, electric flux $\Phi_E = 0$ even if electric fields puncture the surface. Charges outside the Gaussian surface have an equal number of field lines that puncture into and out of the surface.

A **Gaussian surface** is a closed surface.

We find an equation for flux as a function of the charge enclosed in a Gaussian surface. Consider a point charge:

1. Construct a spherical Gaussian surface of radius r around the charge $+q$.
2. Consider a small surface on the sphere with area vector $d\mathbf{A}$. Let $d\mathbf{E}$ be the electric field at the surface.

3. Using equations for electric flux:

$$\Phi_E = EA \cos \theta = \frac{kQ}{r^2} (4\pi r^2) = \frac{1}{4\pi\epsilon_0} (4\pi Q) = \boxed{\frac{Q_{\text{enc}}}{\epsilon_0}}$$

“enc” represents the charge enclosed by the Gaussian surface.

Point charge in Gaussian surface

The electric flux of a charge Q_{enc} enclosed in any Gaussian surface is:

$$\Phi_E = \frac{Q_{\text{enc}}}{\epsilon_0}$$

where:

Φ_E = electric flux

Q_{enc} = enclosed charge (C)

ϵ_0 = vacuum permittivity (C^2/Nm^2)

$= 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$

Surface and volume charge density

$$\sigma = \frac{Q}{A} \quad \rho = \frac{Q}{V}$$

where:

Q = total charge (C)

A = area (m^2)

V = volume (m^3)

σ = surface charge density (C/m^2)

ρ = volume charge density (C/m^3)

Electric field can be found by solving for E in:

$$\Phi_E = \frac{Q_{\text{enc}}}{\epsilon_0} = EA \cos \theta$$

2.10.1 Thin rod

To do

2.10.2 Cylinder

To do

2.10.3 Plane

To do

2.10.4 Capacitor

A capacitor is composed of two conducting and charged planes very close to each other. Net charge is inside the capacitor between the two planes.

We choose a cylinder starting on one plane and ending in between the two planes as the Gaussian surface. Consider one touching the left plane. For the right end cap, area vector is in the same direction as electric field:

$$\Phi_E = EA \cos \theta = EA$$

At the left end cap, electric field is zero because it is inside the plane. Electric field inside a conductor is zero.

$$\Phi_E = \overset{0}{E}A \cos \theta = 0$$

In the sheath, E and A are orthogonal:

$$\Phi_E = EA \cos \theta = 0$$

The total flux for the Gaussian surface is $\Phi_E = EA$. Surface charge density σ is Q_{enc} divided by the area of the end cap: $\sigma = Q_{\text{enc}}/A$. Equating this with $\Phi_E = Q_{\text{enc}}/\epsilon_0$:

$$\Phi_E = \frac{Q_{\text{enc}}}{\epsilon_0} = EA \implies E = \frac{Q_{\text{enc}}}{A\epsilon} = \boxed{\frac{\sigma}{\epsilon_0}}$$

When outside the capacitor, choose a Gaussian surface as a surface partially outside the capacitor and intersecting one plate. $E = 0$ both inside the conducting plane and outside the capacitor, so the total flux $\Phi_E = 0 \implies E = 0$.

2.10.5 Insulating sphere

A charge is uniformly distributed throughout an insulating sphere of radius R .

Inside the sphere. To consider electric field inside the sphere, use a smaller sphere with radius r as the Gaussian surface. At all points on the smaller sphere, the electric field points out normal to its surface. dA is always in the same direction as the electric field.

$$\Phi_E = EA \cos \theta = EA$$

The portion of charge enclosed in the Gaussian sphere is r^3/R^3 .

$$\Phi_E = \frac{Q_{\text{enc}}}{\epsilon_0} = \frac{r^3 Q}{R^3 \epsilon_0}$$

A is the surface area of the sphere: $A = 4\pi r^2$. Combining these:

$$EA = \frac{r^3 Q}{R^3 \epsilon_0} \implies E = \frac{r^3 Q}{(R^3 \epsilon_0)(4\pi r^2)} = \boxed{\frac{Qr}{4\pi R^3 \epsilon_0} = \frac{kQr}{R^3}}$$

The net electric field inside the sphere is directly proportional to the radius of the smaller sphere. When $r = R$:

$$E = \frac{kQR}{R^3} = \frac{kQ}{R^2}$$

is the electric field due to a point charge.

Outside the sphere. Choose a larger sphere of radius r as the Gaussian surface. As before, unit vector is in the same direction as the electric field. A is the surface area of the Gaussian sphere: $A = 4\pi r^2$.

$$\left\{ \begin{array}{l} \Phi_E = EA \cos \theta = EA \\ \Phi_E = \frac{Q_{\text{enc}}}{\epsilon_0} = \frac{Q}{\epsilon_0} \end{array} \right\} \implies E = \frac{Q}{4\pi r^2 \epsilon_0} = \boxed{\frac{kQ}{r^2}}$$

2.10.6 Shell with thickness

The shell has inner radius R_A and outer radius R_B .

The shell is initially neutral. A charge $+q$ is added to the center of the shell (in the cavity); the inside of the shell becomes negatively charged, and since the shell remains neutral, the outside becomes positively charged.

In all cases, dA is parallel to E : $\cos \theta = 1$.

Inside the cavity ($0 < r < R_A$). Only the point charge $+q$ is enclosed: $Q_{\text{enc}} = q$.

$$\left\{ \begin{array}{l} \Phi_E = EA \cos \theta = EA \\ \Phi_E = \frac{Q_{\text{enc}}}{\epsilon_0} = \frac{q}{\epsilon_0} \end{array} \right\} \Rightarrow E = \frac{q}{4\pi r^2 \epsilon_0} = \boxed{\frac{kq}{r^2}}$$

Inside the surface ($R_A < r < R_B$). Negative charges are located at the inner surface (R_A), with magnitude equal to q . So the enclosed charge $Q_{\text{enc}} = 0$.

$$\left\{ \begin{array}{l} \Phi_E = EA \cos \theta = EA \\ \Phi_E = \frac{Q_{\text{enc}}}{\epsilon_0} = 0 \end{array} \right\} \Rightarrow E = \boxed{0}$$

Outside the shell ($r \geq R_B$). The point charge and the net neutrally charged shell is enclosed. The charge of the shell is 0. So the total enclosed charge is $Q_{\text{enc}} = q$.

$$E = \frac{Q}{4\pi r^2 \epsilon_0} = \boxed{\frac{kq}{r^2}}$$

Graph of E against r . E is proportional to $1/r^2$ at all points except when $R_A < r < R_B$, where $E = 0$.

$$E = \begin{cases} 0 & R_A < r < R_B \\ kq/r^2 & 0 < r < R_A \text{ or } r > R_B \end{cases}$$

2.11 Irregularly Shaped Objects in Electrostatic Equilibrium

When an object has a “weird” shape, charges are not distributed uniformly on it. Electrons want to be far apart from each other.

- When the radius of curvature is large (*The Lord Amar of Hightstowne* reference), electrons are able to move a higher distance apart, resulting in lower surface charge density σ and therefore a weaker electric field E .
- When the radius of curvature is low, electrons are not able to move a higher distance apart (there is no component of force pushing them along the surface, and the charge cannot leave the surface), resulting in higher surface charge density σ and therefore a stronger electric field E .

2.12 Electrostatic Equilibrium

An object is in **electrostatic equilibrium** if it has a net charge, but the charges on it are not moving.

When an object is in electrostatic equilibrium, there are no charges moving in or on the surface. There is no net electric field inside the surface: $E = 0$. When inside or on the surface, electric potential is a constant and potential difference at all points is $\Delta V = 0$.

Net charge

- All net charge reside on the surface, as a result of Coulomb's law.
- Electric field "just outside" the surface is perpendicular to the surface.

2.13 Electric Field, Charge and Potential

Inside a conductor

When inside a conductor (both hollow and solid) at electrostatic equilibrium, the electric field is always 0 and electric potential is constant.

Electric field close to conducting surface

Electric field at a location close to a conducting surface with surface charge density σ is:

$$E = \frac{\sigma}{\epsilon_0}$$

where:

E = electric field (N/C)

σ = surface charge density (C/m²)

ϵ_0 = permittivity of free space (C²/Nm²)

$$= 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

This especially works for capacitors.

Net charge and potential

Net charge moves from an area of high potential to an area of low potential.

$$V_1 = \frac{kQ_1}{r_1} \quad V_2 = \frac{kQ_2}{r_2} \quad V_1 > V_2$$

Net charge will flow from area 1 to area 2 until potentials are equal ($V_1 = V_2$):

$$\frac{Q_1}{r_1} = \frac{Q_2}{r_2}$$

Potential and electric field

$$E = -\frac{dV}{dx} \quad \Delta V = -\int E dx$$

where:

E = electric field (N/C = V/m)

V = electric potential (V)

x = distance (m)

- Positively charged objects move from high to low potential.
- Negatively charged objects move from low to high potential.

Exactly the same relationship between force F and potential energy E from mechanics. Objects attempt to minimize their potential/potential energy (not its magnitude).

Example: Potential and electric field of point charge

The electric field created by a point charge Q at a distance r is:

$$E = \frac{kQ}{r^2}$$

Potential difference is between a point with no potential (infinity) and the point of interest r .

$$\Delta V = - \int_{\infty}^r \frac{kQ}{r^2} dr = -kQ \left[-\frac{1}{r} \right]_{\infty}^r = kQ \left(\frac{1}{r} - 0 \right) = \frac{kQ}{r}$$

The derivations below are left as an exercise to the reader.

2.13.1 Arc

The electric potential of an arc is:

$$V = k\lambda \left[\theta \right]_{\theta_1}^{\theta_2}$$

where λ is the linear charge density and θ_1, θ_2 are the bounds of the arc. For a full circle:

$$V = k\lambda(2\pi - 0) = 2\pi k\lambda = 2\pi k \left(\frac{Q}{L} \right) = 2\pi k \left(\frac{Q}{2\pi R} \right) = \frac{kQ}{R}$$

2.13.2 Ring

$$V = \int_0^{2\pi} \frac{k}{d} dq = \frac{k\lambda R}{d} \int_0^{2\pi} d\theta = \boxed{\frac{kQ}{d}}$$

where $\lambda = Q/L = Q/2\pi R$ is linear charge density and $d = \sqrt{R^2 + x^2}$ is diagonal distance to the charge section.

$$V = \boxed{\frac{kQ}{\sqrt{R^2 + x^2}}}$$

where x is the horizontal distance between the point of interest and the ring.

2.13.3 Finite rod

The electric potential due to a finite rod of length L is at a distance a is:

$$V = \int_0^L \frac{k\lambda}{L+a-x} dx = k\lambda \left[-\ln(L+a-x) \right]_0^L = k\lambda (-\ln(a) + \ln(L+a)) = \boxed{k\lambda \ln \left(\frac{L+a}{a} \right)}$$

2.13.4 At perpendicular bisector of finite rod

The electric potential at a distance D from the center of the rod of length L is:

$$V = k \int_{-L/2}^{L/2} \frac{\lambda}{\sqrt{x^2 + D^2}} dx = k \int_0^L \frac{\lambda}{\sqrt{(x-L/2)^2 + D^2}} dx$$

2.14 Superposition

Superposition

The electric potential at a point due to multiple point charges is equal to the sum of the electric potential due to individual charges.

$$V_{\text{total}} = \sum V_i$$

2.15 Electrostatic shielding

A neutral conducting shell will maintain a net charge of zero and electric field $E = 0$, even when placed close to a charge. The shell will be polarized but its net charge remains zero. This is a **Faraday cage**.

3 Capacitors

A **capacitor** consists of two neutral conductors separated by a distance d and a **dielectric** (insulator).

When a capacitor is neutral, it has the same number of positive and negative charges on both plates. The potential difference ΔV across the plates is zero.

When the capacitor is charged, electrons are moved between the plates. Electrons move in the direction the battery's emf makes them move.

As the potential difference across the capacitor increases, it becomes harder to move charges across the capacitor. Charging is complete when the battery's potential difference equals the potential difference across the capacitor. In the end, there is a $+Q$ charge on one plate and a $-Q$ charge on the other.

$$\Delta V_{\text{battery}} = \Delta V_{\text{capacitor}}$$

3.1 Capacitors in Circuits

In a circuit with a battery, resistor and capacitor wired in series:

- **$\Delta V_{\text{capacitor}}$ vs time graph.** When charging, $\Delta V_{\text{capacitor}}$ increases with an asymptote at $\Delta V_{\text{battery}}$ and is concave down. When discharging, it decreases with an asymptote at 0 and is concave up.
- **Current vs time graph.** When both charging and discharging, current decreases and is concave up.
- **Net charge vs time graph.** Same/directly proportional to $\Delta V_{\text{capacitor}}$ vs time graph.
- **$\Delta V_{\text{resistor}}$ vs time graph.** Same/directly proportional to $\Delta V_{\text{capacitor}}$ vs time graph.

Each graph is exponential.

The time it takes to charge the capacitor to potential difference $\Delta V_{\text{battery}}$ depends on:

- The current through the circuit and therefore the resistance of the resistor.
- The area of the capacitor (greater area means there are more charges on the plates).

Energy is required to move charges from one conductor (often a metal plate) to another.

Since the two plates are both neutral initially (with net charge 0 C), at any time, their net charges are equal in magnitude. Let $|Q|$ be the magnitude of the net charge.

3.2 Charging

- At time $t = 0$, both plates are neutral and $|Q| = 0$. The capacitor is equivalent to a closed switch/wire.
- At time $t = \infty$, the potential difference of the capacitor is equal to that of the battery. The capacitor is equivalent to an open switch.

A capacitor is not a battery:

- Both provide a potential difference ΔV and have positive and negative terminals.
- Both store energy, but a battery stores chemical potential energy, while a capacitor contains an electric field and therefore stores electric potential energy.
- Difference: a battery provides a (approximately) constant ΔV . The ΔV and current of a capacitor changes.

3.3 Capacitance

Capacitance

Capacitance is the ability of an object to store charge. It is the slope of a graph of $\Delta V_{\text{capacitor}}$ against the magnitude of charge $|Q|$:

$$C = \frac{\Delta |Q|}{\Delta |\Delta V_{\text{capacitor}}|} = \frac{Q}{\Delta V}$$

where:

Q = charge (C)

ΔV = potential difference (V)

Capacitance is fixed by the geometry of the capacitor. The unit for capacitance is the **farad** (F), which is defined as a coulombs per volt (C/V).

Capacitance is also given by:

$$C = \frac{\kappa \epsilon_0 A}{d}$$

where:

κ = dielectric constant (unitless)

ϵ_0 = permittivity of free space (C^2/Nm^2)

$$= 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

A = area (m^2)

d = distance between plates (m)

The dielectric constant depends on the material used to insulate the two plates (which is called the dielectric).

Recall that $\Delta U_E = q\Delta V$:

Energy of capacitor

The area under the $\Delta V_{\text{capacitor}}$ vs $|Q|$ graph is the energy stored by the capacitor:

$$U_c = \frac{1}{2}Q\Delta V = \frac{Q^2}{2C} = \frac{1}{2}C(\Delta V)^2$$

where:

U_c = energy stored by capacitor (J)

Q = charge (C)

ΔV = potential difference (V)

C = capacitance (F)

Dielectric constant

$$\kappa = \frac{\varepsilon}{\varepsilon_0}$$

where:

κ = dielectric constant (unitless)

ε = permittivity of dielectric material (C^2/Nm^2)

ε_0 = permittivity of free space (C^2/Nm^2)

$$= 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

3.4 RC Circuit

An **RC circuit** is a circuit with a resistor and a capacitor.

An **equivalent capacitor** is a capacitor that has equal capacitance, total charge and potential difference as the original circuit.

Capacitors in parallel

$$C_{\text{eq}} = \sum_i C_i$$

$$\Delta V_{\text{eq}} = \Delta V_i$$

$$Q_{\text{eq}} = \sum_i Q_i$$

where:

C = capacitance (F)

Q = charge (C)

ΔV = potential difference (V)

$$\frac{1}{C_{\text{eq}}} = \sum_i \frac{1}{C_i}$$

$$\Delta V_{\text{eq}} = \sum_i \Delta V_i$$

$$Q_{\text{eq}} = Q_i$$

where:

C = capacitance (F)

Q = charge (C)

ΔV = potential difference (V)

The charge of each capacitor Q_i are equal, because the wires and plates in the middle are shielded by the outermost plates.

Charge can be thought of as current and ΔV as ΔV in a regular circuit. The relationship of individual charges and potential differences are the same as with circuits.

Charge splits at junctions at a ratio equal to C_{eq}/C_i , similar to currents in circuits.

3.5 Distance between Plates

In an RC circuit, ΔV is constant, but in an isolated capacitor, Q is constant.

In both cases, the electric field between the two plates is:

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A} \quad \Delta V = Ed$$

In an RC circuit, ΔV is constant, so electric field strength E decreases. In an isolated capacitor, Q is constant so $E = Q/\epsilon_0 A$ is constant, so ΔV increases with distance.

In an isolated capacitor, charge remains constant w.r.t. distance between plates.

In a capacitor connected to a battery, potential difference remains constant w.r.t. distance between plates.

3.6 Charge Redistribution

When two capacitors are connected, charges will redistribute between the two capacitors. When redistribution is complete:

$$\Delta V_1 = \Delta V_2 \implies \frac{Q'_1}{C_1} = \frac{Q'_2}{C_2}$$

where ΔV is potential difference, Q' is charge after redistribution and C is capacitance. The total amount of charge does not change. If where Q_i are the charges before redistribution:

$$\sum Q_i = \sum Q'_i$$

Potential energy $U_c = Q\Delta V/2$ decreases during redistribution, as energy is dissipated as thermal energy.

3.7 Kirchoff's Rules

A capacitor provides a potential difference that is counted when using Kirchoff's loop rule. Potential drops from the positively charged plate to the negatively charged plate:

$$|\Delta V| = \frac{Q}{C}$$

RC circuit charging

Suppose a battery of potential difference ΔV_1 is connected to a resistor with resistance R and a capacitor. Then the charge Q on each plate of the capacitor is given by the differential equation:

$$\frac{Q_0}{RC} - \frac{dQ}{dt} - \frac{Q}{RC} = 0$$

Solving the differential equation:

$$\frac{t}{RC} = \ln(Q - Q_0) - \ln Q$$

$$Q = Q_0 \cdot \left(1 - \exp\left(\frac{-t}{RC}\right) \right)$$

$$I = \frac{dQ}{dt} = \underbrace{\frac{Q_0}{RC}}_{I_0} \exp\left(\frac{-t}{RC}\right)$$

where:

Q_0 = max value of Q (when capacitor is fully charged) (C)

Q = charge on capacitor at time t (C)

$I = \frac{dQ}{dt}$ = current/rate of change of Q (A)

C = capacitance (F)

t = time (s)

When a capacitor is discharging, a capacitor with potential difference ΔV_C is connected to a resistor with resistance R .

Since the circuit is discharging and Q is decreasing, current is the rate of discharge:

$$I = -\frac{dQ}{dt}$$

RC circuit discharging

The charge Q at time t is given by the differential equation:

$$\Delta V_C - IR = 0 \implies \frac{Q}{RC} + \frac{dQ}{dt} = 0$$

Gives the final result:

$$Q = Q_0 \exp\left(\frac{-t}{RC}\right)$$
$$I = -\frac{dQ}{dt} = \underbrace{\frac{Q_0}{RC}}_{I_0} \exp\left(\frac{-t}{RC}\right)$$

3.8 Time Constant

Time constant

The **time constant** (τ) of an RC circuit is defined as the product of resistance and capacitance:

$$\tau = RC$$

where:

τ = time constant (s)

R = resistance (Ω)

C = capacitance (F)

The product RC has units of seconds. From $C = Q/\Delta V$, a farad is a coulomb per volt (C/V) and from $R = V/I$, an ohm is a volt per ampere (V/A).

$$\left[\text{F} \cdot \Omega = \frac{\text{C}}{\text{V}} \cdot \frac{\text{V}}{\text{A}} = \frac{\text{C}}{\text{A}} = \frac{\text{C}}{\text{C/s}} = \text{s} \right]$$

Using the capacitor charging and discharging equations:

$$Q = Q_0 \cdot \left(1 - \exp\left(\frac{-t}{RC}\right) \right)$$

$$Q = Q_0 \exp\left(\frac{-t}{RC}\right)$$

Charging. At time $t = \tau = RC$, the charge on a capacitor is:

$$Q = Q_0 \cdot \left(1 - \exp\left(\frac{-RC}{RC}\right) \right) = Q_0(1 - e^{-1}) \approx 0.632Q_0$$

At the time constant, the capacitor is approximately 63.2% charged.

Discharging. At time $t = \tau = RC$, the charge on a capacitor is:

$$Q = Q_0 \exp\left(\frac{-RC}{RC}\right) = Q_0 e^{-1} \approx 0.368Q_0$$

At the time constant, the capacitor is approximately 36.8% charged. If the capacitor was full at $t = 0$, it has lost 63.2% of charge.

Time constant with multiple resistors and capacitors

$$\tau = R_{\text{eq}} C_{\text{eq}}$$

where:

τ = time constant (s)

R_{eq} = equivalent resistance (Ω)

C_{eq} = equivalent capacitance (F)

3.9 Additional Circuit Equations

RC circuits are transient circuits, meaning the behavior (potential differences, currents, etc) of the circuit changes over time. Circuits done previously in part 1 where $I, \Delta V_B, R, P$ are constant are called **steady state circuits**.

Since I is flow rate of charge, the total charge moved through any circuit (including transient circuits) is the integral of current.

Current and charge

$$I = \frac{dQ}{dt}$$
$$\Delta Q = \int I dt$$

where:

I = current (A)

Q = charge (C)

Similarly, power is change in energy over time, so change in energy is the integral of power.

Power and energy

$$P = I\Delta V = \frac{dE}{dt}$$
$$\Delta E = \int P dt$$

where:

P = power (W)

E = energy (J)

3.10 Cylindrical Capacitors

Let r_1 be the inner radius and r_2 be the outer radius. From Gauss's law, the electric field inside a cylindrical capacitor is:

$$E = \frac{Q}{2\pi r L \epsilon_0}$$

where $r_1 \leq r \leq r_2$ is the radius of the Gaussian surface and L is its length. The potential difference the capacitor provides is:

$$\Delta V = - \int_a^b E \, dr = \frac{Q}{2\pi L \epsilon_0} \ln \frac{b}{a}$$

This integral is evaluated from areas of low potential to areas of high potential. The capacitance is:

$$C = \frac{Q}{\Delta V} = \frac{Q}{\frac{Q}{2\pi L \epsilon_0} \ln(a/b)} = \frac{2\pi L \epsilon_0}{\ln(b/a)}$$

The capacitance depends only on geometry and not on the charge on the plates Q . If the order of bounds of the integral for ΔV is switched, it is important to ensure C is a positive value.

To charge a cylindrical capacitor:

1. Start with neutral cylindrical shell.
2. Charge inner cylinder with positive charges.
3. Connect outer shell to ground, causing negative charges to move onto it.

3.11 Spherical Capacitors

The electric field between the outer and inner “plates” is, using a Gaussian surface of radius r :

$$E = \frac{Q}{4\pi r^2 \epsilon_0}$$

The potential difference between two distances r is:

$$\Delta V = - \int_a^b E \, dr = \frac{Q}{4\pi \epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$$

The capacitance is:

$$C = \frac{Q}{\Delta V} = 4\pi \epsilon_0 \left(\frac{1}{a} - \frac{1}{b} \right)$$

3.12 Dielectrics

The dielectric is an insulator placed in the space between the plates of a capacitor.

$$C = \frac{\kappa \epsilon_0 A}{d}$$

A more insulating dielectric increases the dielectric constant (relative permittivity) κ , increasing the capacitance.

The dielectric also prevents contact between the plates, which would result in a short circuit.

Isolated capacitor. The side of the dielectric facing the negative side becomes slightly positive and the side facing the positive side becomes slightly negative.

The dielectric is polarized by the capacitor plates, and has an electric field of its own antiparallel to that of the capacitor. It is much weaker, so the electric field of the capacitor is weakened (fewer electric field lines).

Different materials polarize more/less easily, resulting in a different change in electric field strength. The dielectric constant κ changes.

$$C = \frac{\kappa \epsilon_0 A}{d} = \frac{Q}{\Delta V}$$

Higher dielectric constant κ results in higher capacitance and lower potential difference (lower electric field).

Doing work on the capacitor (such as inserting a dielectric in between it) increases potential energy U_c .

Dielectric

$$\kappa = \frac{C_d}{C_0}$$

where:

κ = dielectric constant (unitless)

C_d = capacitance with dielectric (F)

C_0 = capacitance without dielectric (F)

This can be derived from the equation for capacitance $C = \kappa\epsilon_0 A/d$.

In an isolated capacitor, charge remains the same. The dielectric constant is a ratio of capacitance. It can also be expressed in terms of electric field and potential difference:

$$\kappa = \frac{Q/\Delta V_d}{Q/\Delta V_0} = \frac{\Delta V_0}{\Delta V_d} = \frac{E_0}{E_d}$$

Connected to circuit. Adding a dielectric will initially reduce its E and ΔV . A capacitor connected to a battery maintains its ΔV as more charges are pumped to the plates. The original E and ΔV are reestablished.

Q , κ and C will increase, and potential energy U_c increases. d , A , ΔV and E are constant.

A capacitor stores more energy for a given ΔV when it has higher capacitance/with a dielectric (In an isolated capacitor, ΔV is not fixed). The dielectric is attracted when inserted as energy is drawn from the battery. Removing the dielectric causes the capacitor to discharge (decrease in capacitance) and recharges the battery.

Increasing κ increases the slope of a ΔV vs Q graph.

3.12.1 Partially Filled

A dielectric that only fills a fraction of the space between two capacitor plates.

Case 1. Half of the capacitor is filled with a dielectric parallel to the plates, the other half is a vacuum. This is equivalent to two capacitors connected in series: one filled by the dielectric material, the other a vacuum with proportion, each with distance $d/2$. The capacitances are C_1 for the vacuum, C_2 for the dielectric:

$$C_2 = \frac{k\epsilon_0 A}{d/2} = 2\kappa C_0$$

$$C_1 = \frac{\epsilon_0 A}{d/2} = 2C_0$$

The equivalent capacitance is:

$$C_{eq} = \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1} = \left(\frac{2k}{k+1} \right) C_0$$

where C_0 is the capacitance over a vacuum with area A and distance d .

$$C_0 = \frac{\epsilon_0 A}{d}$$

Case 2. Half of the capacitor is filled orthogonal to the plates. This is the same as case 1 but the two capacitors are connected in parallel.

$$C_{\text{eq}} = C_1 + C_2 = \left(\frac{1 + \kappa}{2} \right) C_0$$

Case 3. The capacitor is filled by a metal slab of thickness a . Since the metal is a conductor charges will completely separate to the surface. This is equivalent to only two capacitors, both filled with vacuums; the metal plate becomes just a wire. The distances for both capacitors is half the space without the metal plate: $(d - a)/2$.

$$C_1 = C_2 = \frac{\epsilon_0 A}{(d - a)/2} = \frac{2\epsilon_0 A}{d - a}$$

$$C_{\text{eq}} = \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1} = \frac{\epsilon_0 A}{d - a}$$

4 Magnetism

Moving charges (current or individual charges moving) create magnetic fields. There are charges moving in permanent magnets as well.

While electric charges can be isolated, magnetic monopoles do not exist (always a north and south pole).

Outside the magnet, magnetic field lines point from the north to the south pole. There are magnetic field lines inside the material as well in such a way to make them closed loops (this means that inside a bar magnet, the lines are from south to north). The symbol B refers to a magnetic field. In a simplified field diagram:

- Field lines are parallel just outside the material.
- Sometimes there is a loop from the north to south pole as well.

To draw magnetic field lines in 3D: “X”’s indicate a magnetic field directly into the page, while dots indicate one directly out of the page.

When a magnet is dropped inside a magnetic field, it will orient itself such that the magnetic field lines are aligned/pointing in the same direction.

Magnetic force

$$F_B = q \cdot \mathbf{v} \times \mathbf{B} = |q| v B \sin \theta$$

where:

$F_B = F_M$ = magnitude of magnetic force on charged particle (N)

q = charge (C)

$\mathbf{v}$ = velocity of charged particle (m/s)

$\mathbf{B}$ = magnetic field (T)

$v = \|\mathbf{v}\|$ = speed of charged particle (m/s)

$B = \|\mathbf{B}\|$ = magnitude of magnetic field (T)

An uncharged or stationary particle is not affected by magnetic force: $q = 0, v = 0 \implies F_B = 0 \text{ N}$. If the particle moves parallel/antiparallel to magnetic field lines, then $\sin \theta = 0 \implies F_B = 0 \text{ N}$. Magnitude is maximized when $\theta = 90^\circ \implies \sin \theta = 1$.

A negative charge will move in the opposite/left handed direction $-q \cdot \mathbf{v} \times \mathbf{B} = q \cdot \mathbf{B} \times \mathbf{v}$.

The unit for magnetic field is the tesla (T), which is equivalent to:

$$\left[\text{T} = \frac{\text{N}}{\text{C} \cdot \text{m/s}} = \frac{\text{N}}{(\text{A} \cdot \text{s}) \cdot \text{m/s}} = \frac{\text{N}}{\text{A} \cdot \text{m}} \right]$$

Charged particles move in magnetic fields. Charged particles include:

- Proton (p)
- Electron (e^-, β^-)
- Positron (electron with positive charge e^+, β^+)
- Neutron (n)
- Helium nucleus (${}^4_2\text{He}$)

Protons, electrons, positrons have magnitude of the fundamental charge $\pm 1.6 \times 10^{-19} \text{ C}$. The helium nucleus has twice the charge $+3.2 \times 10^{-19} \text{ C}$ while neutrons are neutral. A positron has the mass of an electron but the charge of a proton.

4.1 Applications of Magnetic Fields

Deflection of charges by magnetic fields. To determine the direction of movement of charged particles, use the right hand rule. The direction of magnetic force is orthogonal to both the magnetic field B and the velocity of $+$ charges v , and is parallel to $v \times B$. Charges move in a circular arc; F_B (force from magnetic field) changes the direction of particles but not the speed.

When the magnetic force is not perpendicular with the velocity, the particle moves in a helix.

Charged particles in magnetic fields. When a charge is placed in a magnetic field such that its velocity is

perpendicular to magnetic field lines, it moves in uniform circular motion (at constant speed). In this case:

$$\Sigma F = ma \implies q \cdot \mathbf{v} \times \mathbf{B} = qvB \sin \theta = \frac{mv^2}{r} \implies \boxed{qB = \frac{mv}{r}}$$

Mass spectrometer/spectrograph. There are usually four parts of a mass spectrometer:

1. A molecular sample is injected and vaporized.
2. The molecules are ionized and obtain a (usually +1) charge.
3. Linear accelerator: ionized particles are accelerated to high speeds.
4. There is a magnetic force here that deflects the particles towards a deflector. The mass of the ions change the position on the detector they hit, where distance $r = mv/qB$.

4.2 Magnetism due to Wires

Direction and magnitude of magnetic field due to wires

The magnetic field around a wire are concentric circles. The direction of the circular magnetic fields are given by the curled fingers of the right hand where the thumb is in the direction of conventional current (opposite of electron flow).

$$B = \frac{\mu_0 I}{2\pi r}$$

where:

B = magnitude of magnetic field (T)

μ_0 = vacuum permeability (Tm/A)

$= 4\pi \times 10^{-7}$ Tm/A

I = current (A)

r = perpendicular distance from wire (m)

Magnetic force on current-carrying wire

The force exerted on a current-carrying wire by a magnetic field is:

$$\mathbf{F}_B = l \cdot \mathbf{I} \times \mathbf{B}$$

where:

$\mathbf{F}_B$ = magnetic force (N)

$\mathbf{I}$ = current (A)

l = length of wire in magnetic field (m)

$\mathbf{B}$ = magnetic field (T)

The direction of the force can be determined by the right hand rule. $\mathbf{I}$ is a vector pointing in the direction of conventional current, and length l is a scalar.

Force between two parallel wires

The magnitude of force exerted by two parallel current-carrying wires on each other is:

$$F_B = \frac{\mu_0 I_1 I_2 l}{2\pi r}$$

where:

F_B = magnetic force (N)

I_1 = current in wire 1 (A)

I_2 = current in wire 2 (A)

l = length of wires (m)

r = distance between wires (m)

μ_0 = vacuum permeability (Tm/A)

$$= 4\pi \times 10^{-7} \text{ Tm/A}$$

Wires with parallel currents attract each other while wires with anti-parallel currents repel.

Magnetic field at center of current-carrying coils

The magnetic field at the center of a coil/loop is:

$$B = \frac{\mu_0 I}{2r}$$

where:

B = magnetic field (T)

I = current in coil (A)

r = radius of coil (m)

μ_0 = vacuum permeability (Tm/A)

$$= 4\pi \times 10^{-7} \text{ Tm/A}$$

Solenoid

A solenoid (multiple coils of wire) behaves as an electromagnet: a magnet where the north pole is the end where conventional current enters (“+”), and the south pole is where conventional current exits (“-”). The strength of the magnetic field is:

$$B = \mu_0 n I$$

where:

B = magnetic field (T)

n = number of coils per unit length (m^{-1})

I = current (A)

μ_0 = vacuum permeability (Tm/A)

$$= 4\pi \times 10^{-7} \text{ Tm/A}$$

4.3 Electromagnetic Induction

Induction is used in:

- Generators and motors (mechanical energy $\longleftrightarrow$ electrical energy).
- Microphones and speakers (sound energy $\longleftrightarrow$ electrical energy).
- Transformers, rail guns and oven/cooktop/“induction stoves.”

Potential difference induced by magnetic field. When a neutral conductor is placed in a magnetic field, it is polarized as electrons are moved by $F_B = q \cdot \mathbf{v} \times \mathbf{B}$. The polarization leads to an electric force F_E pointing in the opposite direction to F_B . F_E and F_B will equilibrate.

$$F_E = F_B \implies qE = q \cdot \mathbf{v} \times \mathbf{B} \implies E = \mathbf{v} \times \mathbf{B}$$

The potential difference between the polarized ends of the conductor is:

$$\Delta V = Ed = l \cdot \mathbf{v} \times \mathbf{B} = Blv$$

where $\Delta V = \varepsilon$ is the potential difference established across conductor as a result of relative movement between a conductor and the magnetic field, and l is the length of the conductor. B is the magnitude (strength) of the electric field: $B = \|\mathbf{B}\|$. v is the relative speed between the conductor and the magnetic field. The latest equality $\Delta V = Blv$ is only true when velocity v and magnetic field B are orthogonal: $\|\mathbf{v} \times \mathbf{B}\| = vB$.

Relative motion between a conductor and a magnetic field induces a potential difference in the conductor.

Rail system. A rail system consists of a conductor, a magnetic field and two **rails**. The rails are attached to the conductor and joined by a resistor with resistance R . The rails are stationary and conductive.

The potential difference induced by the magnetic field creates a current through the rails and the resistor.

This system is a generator: the movement/mechanical energy of the conductor through the magnetic field induces a potential difference and generates an electric current.

$$I = \frac{\Delta V}{R} = \frac{Blv}{R}$$

The induced current flows in a direction that opposes the motion of the rod and magnetic force. The magnetic force on the current in the wire has magnitude $F_B = IlB$.

Lenz's law

Induced current always flows in such a way to oppose the change that caused it.

Example: Lenz's law

For example, a wire loop is moved out of a magnetic field. This decreases the flux in a certain direction, which induces a potential difference $|\Delta V| = |d\Phi/dt|$. The direction the current flows is in the direction that resists the decrease in flux (i.e. the current causes the flux to increase) by the right hand rule.

Recall from power $P = Fv$, the power of the applied force (force moving the conductor across the field) is:

$$P = F_A v = F_B v = (IlB)v = \frac{Blv}{R}(lB)v = \frac{B^2 l^2 v^2}{R}$$

By conservation of energy, applied force equals magnetic force: $F_A = F_B$. The power of the electrical current through the rails is:

$$P = I^2 R = \left(\frac{Blv}{R} \right)^2 R = \frac{B^2 l^2 v^2}{R}$$

Magnetic flux

$$\Phi_B = \oint B \cdot dA = BA \cos \theta$$

where:

Φ_B = magnetic flux ($\text{Tm}^2 = \text{Vs}$)

B = magnitude of magnetic field (T)

A = area (m^2)

θ = angle between area vector and magnetic field

Faraday's law. Monitor the magnetic flux Φ_B inside the loop of a rail system.

$$\Phi_B = BA \cos \theta = BA$$

A is the area of the current loop: through the rail, into the resistor, through the other rail, and back through the conductor. The directed area vector of this loop is parallel to the magnetic field, so $\theta = 0$. As time increases, the area bound by the loop increases.

The change in flux with respect to time is:

$$\frac{\Delta \Phi_B}{\Delta t} = \frac{\Delta(Bls)}{\Delta t} = Bl \cdot \frac{\Delta s}{\Delta t} = Blv = \Delta V$$

where s is the length/width of the loop (the distance between the resistor and the conductor).

Faraday's law

$$\Delta V = -N \cdot \frac{d\Phi_B}{dt}$$

N is the number of turns in the loop.

4.4 Biot-Savart Law

Biot-Savart law

The Biot-Savart law is used to find the magnetic field caused by a constant current.

$$d\vec{B} = \frac{\mu_0}{4\pi} \cdot I \cdot \frac{d\vec{l} \times \hat{r}}{r^2}$$
$$dB = \frac{\mu_0}{4\pi} \cdot I \cdot \frac{dl \sin \theta}{r^2}$$

where:

I = current (A)

dl = small wire segment in the direction of current

$\hat{r}$ = unit vector from wire segment to point of interest

r = distance from wire segment to point of interest (m)

θ = angle between dl and $\hat{r}$

μ_0 = vacuum permeability (Tm/A)

$$= 4\pi \times 10^{-7} \text{ Tm/A}$$

5 Induction

An **inductor** is a “little” solenoid. When current flows, a magnetic field is established in the inductor. If the current is changed, the magnetic flux is changed as well.

When the switch is closed, by Faraday’s law and Lenz’s law, a potential difference is induced in the inductor that opposes the change (of closing the switch) and of the potential difference from the battery. This induced potential difference in the inductor is called “self-induced” or “back emf/ ΔV ”.

Inductors (temporarily) resist changes in current. At time $t = 0$, the current is zero since the inductor acts like a battery with opposite polarity. As time $t \rightarrow \infty$, the current approaches as if there were no inductor: $I = \Delta V/R$, and the current is steady.

The graph of ΔV against time is increasing and concave down.

Energy is stored in the inductor but the inductor does not have any resistance.

In a solenoid, there is a magnetic field in the same direction as current. The flux for a single loop is:

$$\Phi_B = BA \cos \theta = BA$$

If the inductor has n loops, the flux is $n\Phi_B$. This is proportional to current I .

$$N\Phi_B = LI$$

L is the *inductance*.

Inductance

Inductance L is the proportionality constant between the flux of the inductance and current.

$$n\Phi_B = LI \quad L = \frac{n\Phi_B}{I} = \frac{nBA \cos \theta}{I}$$

where:

n = number of loops in inductor

Φ_B = magnetic flux ($\text{T} \cdot \text{m}^2$)

B = magnetic field (B)

A = area (m^2)

I = current (A)

The unit of inductance is the henry (H).

$$\left[\frac{\text{T} \cdot \text{m}^2}{\text{A}} = \text{H} \right]$$

The magnetic field of a solenoid is:

$$B = \mu_0 n I$$

So inductance equals:

$$L = \frac{nBA \cos \theta}{I} = \frac{\mu_0 n^2 A}{l}$$

where n is the number of loops, A is the cross-sectional area and l is the length of the inductor.

Induced potential difference and inductance. By Faraday's law, the change in magnetic flux with respect to time is the potential difference ΔV .

$$LI = n\Phi_B \implies L \cdot \frac{\Delta I}{\Delta t} = n \cdot \frac{\Delta \Phi_B}{\Delta t} \implies \Delta V_{\text{induced}} = -L \cdot \frac{\Delta I}{\Delta t} = -L \cdot \frac{dI}{dt}$$

The negative sign indicates that the induced potential difference opposes the change in current. The faster the current changes, the greater the induced current is. After a long time, the inductor acts just like a wire since the change in current over time approaches zero.

Energy stored in inductor

The magnetic potential energy stored in an inductor is:

$$U_L = \frac{1}{2} LI^2$$

where:

U_L = magnetic potential energy (J)

L = inductance (H)

I = current (I)

5.1 LR Circuits

Recall that in an RC circuit the energy stored in the capacitor is $U_C = Q^2/2C = C(\Delta V)^2/2$. Current decreases/concave up, potential different across the capacitor increases/concave down. Similar for discharging.

A resistor-inductor circuit is like an RC circuit but with an inductor instead. When the resistor and inductor is connected to the battery, the inductor resists the battery's current. This is called **energizing**. Current increases over time.

At time $t = 0$, an inductor acts as a battery acting opposite the battery. As time $t \rightarrow \infty$, the inductor acts as a wire with no resistance. When energizing, current increases/concave down; ΔV across the inductor decreases/concave up.

When **de-energizing**, current decreases over time, because there is a magnetic field inside the inductor's coils. It will provide a current in the same direction as before the switch is moved. Over time, the potential energy stored in the inductor is converted to thermal energy. Current decreases/concave up, ΔV across the inductor increases/concave down.

5.2 LC Circuit

Imagine a circuit with an open switch, charged capacitor and inductor. When the switch is closed, there will be a growing magnetic field in the inductor pointing in the direction of the current given by the capacitor. There is a maximum amount of current flowing through the inductor when the capacitor plates are neutral. Now the inductor will function as a battery and keep pushing current in the same direction, which will charge the capacitor in the opposite direction as it starts (the negative capacitor plate is now positive), until the capacitor is fully charged and there is no magnetic field in the inductor. If there is no resistance, this cycle will continue forever.

The time it takes for the capacitor to discharge and then fully recharge with the same initial polarity (the circuit returns to the exact same state) is the period T .

Period of LC circuit

$$\omega = \frac{1}{\sqrt{LC}} \iff T = 2\pi\sqrt{LC}$$

where:

L = inductance (H)

C = capacitance (T)

ω = angular frequency (rad/s)

T = period (s)

ω is the angular frequency. $\omega = 2\pi f$ where f is frequency, and $T = 1/f$.

LC circuits is equivalent to simple harmonic motion, such as in a spring-mass system.